

LAB 1: Inspecting Table Size

- **Goal:** Identify which tables in dvdrental are "Index Heavy".

	object_name name	data_size text	index_size text	total_size text
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	city	64 kB	48 kB	112 kB
9	film_category	72 kB	40 kB	112 kB
10	actor	40 kB	32 kB	72 kB
11	store	8192 bytes	32 kB	40 kB
12	staff	16 kB	16 kB	32 kB
13	language	8192 bytes	16 kB	24 kB
14	category	8192 bytes	16 kB	24 kB
15	country	8192 bytes	16 kB	24 kB

- Tables are consider "index heavy" when the index_size is larger than data_size or the ratio $\frac{\text{index size}}{\text{total size}} > 0.7$
- From the images, tables *rental* and *payment* are "index heavy"

Lab 2: The Impact of a New Index

- **Discussion:** How many Pages (8KB) did this new index consume?

Before indexing

	pg_size_pretty text
1	920 kB

After indexing

	pg_size_pretty text
1	1488 kB

- The new index consumes 568kB which is 71 Pages.

Lab 3: Partial Index Practice

- **Task:** Compare this size with a hypothetical full index on `replacement_cost`.

	index_name name	index_size text	size_bytes bigint
1	idx_full_replacement_c...	40 kB	40960
2	idx_expensive_films	16 kB	16384

- **Partial index** is smaller because it only save index entry for films with *replacement_cost* > 25, while **Full index** save entry for all films.